

A plume origin for hydrous melt at the lithosphere–asthenosphere boundary

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Plate tectonics requires a low-viscosity layer beneath the lithosphere–asthenosphere boundary (LAB), yet the origin of this ductile transition remains debated^{1,2}.

Explanations include the weakening effects of increasing temperature^{3,4}, mineral hydration⁵ or partial melt⁶. Electrical resistivity is sensitive to all three effects⁷, including melt volatile content⁸, but previous LAB constraints from magnetotelluric soundings did not simultaneously consider the thermodynamic stability of the inferred amount of melt and the effect of uncertainty in the estimated resistivity^{8–14}.

Here we couple an experimentally constrained parameterization of mantle melting in the presence of volatiles^{15,16} with Bayesian resistivity inversion¹⁷ and apply this to magnetotelluric data sensitive to a LAB channel beneath the Cocos Plate⁹.

Paradoxically, we find that the conductive channel requires either anomalously large melt fractions with moderate volatile contents or moderate melt fractions with anomalously large volatile contents, depending on the assumed mantle temperature. Large melt fractions are unlikely to be mechanically stable and conflict with melt-migration models¹⁸. As large volatile contents require a highly enriched mantle source inconsistent with mid-ocean-ridge estimates¹⁹, our results indicate that a mantle plume emplaced volatile-rich melts in the LAB channel. This requires the presence of a previously undetected nearby plume or the influence of the distant Galápagos hotspot. Plumes that feed thin, hydrous melt channels^{9,14,20} may be an unrecognized source of LAB anomalies globally.

The decoupling of sliding lithospheric plates from the deeper convecting mantle at the LAB is a critical component of plate tectonics theory. The bulk mantle properties that govern this decoupling can be analysed by geophysical observations such as the magnetotelluric (MT) method, which uses natural variations in the electric and magnetic fields of the Earth to image mantle resistivity. MT data are primarily sensitive to conductive structures, where large currents are induced. This feature, combined with the dependence of mantle resistivity on bulk hydration, temperature and partial melt, makes MT data useful for investigating the nature of the asthenosphere⁷. Seismic data are also commonly used to image the LAB². Both low resistivity and low seismic velocity can be explained by higher melt fractions at lower temperature or by lower melt fractions at higher temperature (Fig. 1a). Resistivity, however, is also sensitive to the melt volatile content⁸, in which a parcel of mantle with more melt but fewer dissolved volatiles can explain the same resistivity as a parcel with less melt but more dissolved volatiles (Fig. 1b). Although these trade-offs make electric and seismic mantle structure alone insufficient for unambiguously discerning the origin of the asthenosphere, the uncertainty can be notably reduced by requiring that the combination of temperature, melt fraction and volatile concentration be in thermodynamic equilibrium with the surrounding mantle.

Mantle melting increases with temperature (blue curve in Fig. 1a), in inverse correlation with resistivity and seismic velocity. The addition

of volatiles, which lower the peridotite solidus¹⁵, further increases melting. At fixed temperature, the petrologically stable melt fraction and the volatile content correlate positively: the higher the volatile concentration, the higher the stable melt fraction (blue curves in Fig. 1b), whereas for a fixed resistivity, the opposite is true⁸: increasing volatile content requires a lower melt fraction. Because of this, simultaneous knowledge of both resistivity and temperature should uniquely specify the stable melt fraction and volatile content (for example, pink circle in Fig. 1b), key properties that strongly influence the mantle viscosity²¹. Including quantitative uncertainty in mantle resistivity and temperature leads to a region in the melt fraction/volatile content space that represents how well our knowledge of mantle resistivity and temperature can constrain these quantities (shaded region in Fig. 1b).

A previous MT study on 22–23 million years (Myr) old Cocos Plate lithosphere offshore Nicaragua imaged a thin, low-resistivity LAB channel⁹ (Extended Data Fig. 1). Although the conductive channel is consistent with a layer of low-degree partial melt, the available modelling tools did not permit estimation of the uncertainty in the resistivity or, by implication, the amount of partial melt and volatiles at the LAB. Here we re-examine a subset of these data compatible with 1D modelling using new tools for non-linear uncertainty estimation for MT data¹⁷ in conjunction with a parameterization of volatile-influenced silicate melting^{15,16} to place quantitative bounds on the amount of thermodynamically stable melt and volatiles at the LAB beneath the Cocos Plate.

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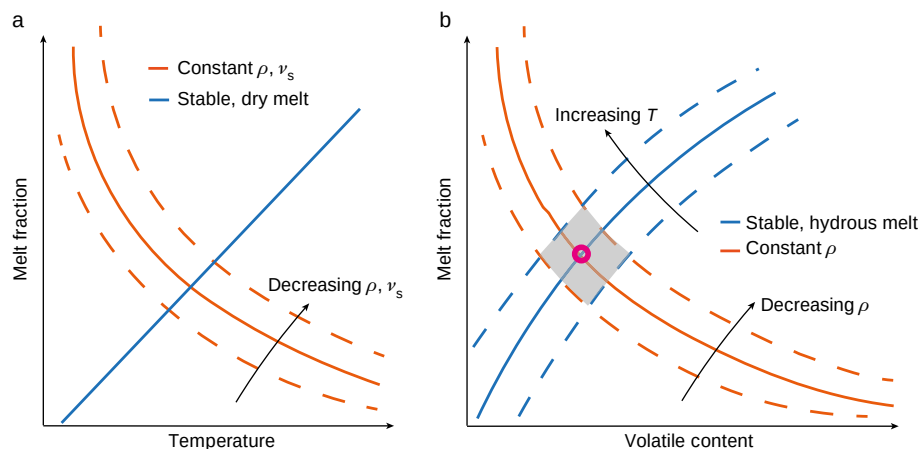


Fig. 1 | Diagrams illustrating trade-offs inherent in estimating mantle melt fraction and volatile content. **a**, Melt fraction trades off with temperature in explaining bulk resistivity ρ and shear wave velocity v_s . **b**, Separately, melt fraction trades off with melt volatile content, to which only resistivity is

sensitive. Blue curve in **a** shows a stable, dry melt curve, whereas those in **b** show stable, hydrous melt curves at constant temperature, allowing uncertainty (dashed lines) in resistivity (orange) and temperature (blue) to constrain mantle melt fraction and melt volatile content (grey region).

Monte Carlo estimates of melt fraction

We estimate probability distributions for melt fraction and melt volatile content (here ‘volatile’ means water and carbon dioxide) from distributions for mantle resistivity and temperature using an iterative Monte Carlo approach. At each iteration, samples for both resistivity (ρ) and temperature (T) are drawn from their respective distributions and the combination of melt fraction and melt volatile content that explains the sampled resistivity and that is thermodynamically stable with respect to the sampled temperature is calculated, using the physical models and relationships outlined in Methods. Assuming that the melt is generated in situ and remains static, we can also infer the corresponding bulk mantle volatile inventory responsible for its formation. Probability distributions for bulk resistivity at the LAB were obtained by Bayesian inversion of a subset of the MT data shown in Extended Data Fig. 1. The full Bayesian posteriors for resistivity as a function of depth for two MT stations are shown in Extended Data Fig. 2. The mantle temperature at 57.5 km depth (the midpoint of the LAB channel) was assumed to be uniformly distributed over 1,216–1,418 °C, on the basis of plate-cooling models at a range of mantle potential temperatures (MPTs) to account for the uncertainty^{22,23}. Details on how these distributions were obtained can be found in Methods.

The results of the Monte Carlo process statistically require melt to explain the conductive anomaly shown beneath the Cocos Plate by the

MT data. Indeed, the amount of mineral hydration required to explain the bulk resistivity with a melt-free mantle always depresses the solidus below the mantle temperature and thus requires melt. This conclusion holds despite the large uncertainty in the resistivity of mantle minerals under hydrous conditions (see Methods). Previous analysis of these data involving gradient-based inversion⁹ indicated that partial melt was probably required, but was unable to statistically quantify the range of compatible melt fractions. Nearly all (>99%) of the melt fractions produced through our Monte Carlo process are greater than zero.

Although there is no ambiguity about the presence of melt, we find a large uncertainty in the amount of melt and volatiles at the LAB. If the bulk CO₂ inventory is fixed at 100 ppm and bulk hydration is allowed to vary to match the (T, ρ) pairs drawn at each iteration of the Monte Carlo process, we obtain the results shown in Fig. 2. Melt fraction ranges from 1 to 10 vol%, whereas melt H₂O concentration varies over 0–6 wt%, with higher melt fractions corresponding to less water. This implies 0–2,000 ppm H₂O in the mantle source. Owing to the 100 ppm fixed bulk CO₂ content, the melt CO₂ content is consistently less than 2 wt%, which is too small to affect the bulk resistivity.

The primary reason for this wide range of melt fraction values is the uncertainty in temperature. Comparing the probability distribution for melt fraction, melt volatile content and bulk mantle hydration for all Monte Carlo samples with marginal distributions for samples taken at discrete temperatures, we observe a clear trade-off between

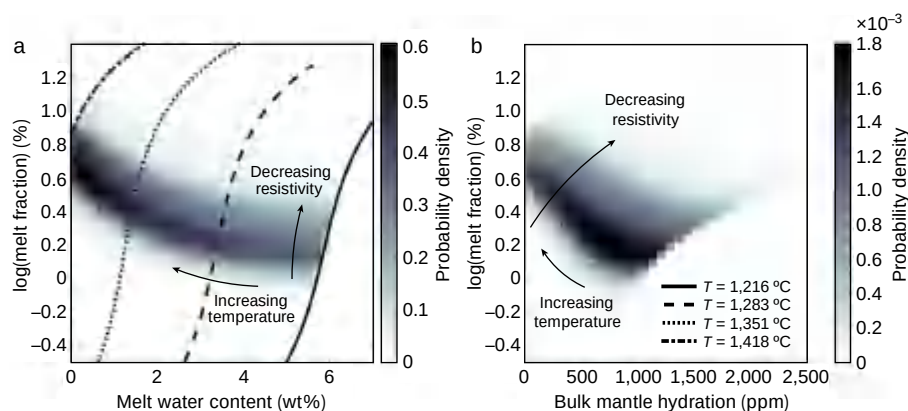


Fig. 2 | Probability distributions for melt fraction and melt H₂O content or bulk mantle hydration obtained from a Monte Carlo method. **a**, Melt H₂O content. **b**, Bulk mantle hydration. Seven MT soundings (sites s03, s04, s06, s08, s09, s10 and s11; see Extended Data Fig. 1) were used. Dashed lines indicate isotherms. Bulk carbon dioxide was held constant at 100 ppm. Nearly all of the (T, ρ) draws require the presence of hydrous melt.

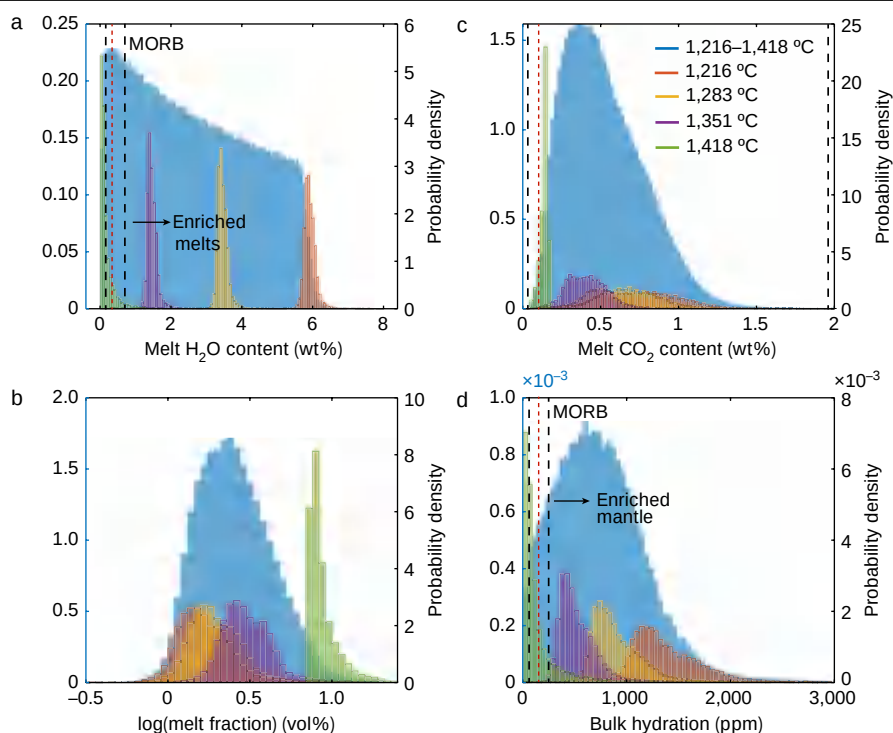


Fig. 3 | Marginal distributions for melt H₂O, melt fraction, melt CO₂, and bulk hydration from a Monte Carlo process. **a**, Melt H₂O. **b**, Melt fraction. **c**, Melt CO₂. **d**, Bulk hydration. The Monte Carlo process uses the resistivity distribution from the MT data at discrete temperatures (orange, yellow, purple and green distributions). The distributions obtained from the full temperature range are plotted behind in blue. The warmest temperatures require less hydration to explain the bulk resistivity, but result in higher melt fraction. Bulk

carbon dioxide inventory was held fixed at 100 ppm. As a result, at no temperature were the melt fractions small enough for melt CO₂ concentrations to exceed 2 wt%. Black and red dashed lines in the upper-left panel represent the 5th, 50th and 95th percentiles, respectively, of melt volatile measurements of MORB glasses¹⁹, whereas those in the lower-right panel represent the mean and 2 σ uncertainty of inferred bulk mantle hydration²⁵. Higher degrees of hydration are considered enriched.

temperature and hydration in explaining the observed resistivity distributions (Fig. 3). Higher temperatures produce larger melt fractions and require less water (green distributions), whereas lower temperatures produce less melt but require higher melt H₂O (orange distributions). If mantle temperature is more accurately known, tighter constraints could be placed on the LAB melt fraction and volatile content.

To evaluate whether the observed resistivity distributions could be explained with very-low-degree ($\ll 1$ vol%) carbon-rich melts⁸, we repeated the Monte Carlo process with bulk mantle hydration fixed at 240 ppm H₂O and bulk mantle CO₂ allowed to vary to match the (T , ρ) pairs drawn at each iteration (Extended Data Fig. 3). Most draws required very high bulk mantle CO₂ concentrations (in excess of 500 ppm) and only for the coldest temperatures (associated with anomalously cold MPTs) were very-low-degree melts stable (Extended Data Fig. 4). Yet such cold MPTs are at odds with anomalously shallow seafloor bathymetry in this region, which suggests that the Cocos Plate is relatively warm for its age²⁴. We therefore interpret the conductive anomaly imaged beneath the Cocos Plate to be a volatile-rich partial melt channel containing moderate melt fractions (1–3 vol%).

Possible origins for hydrous melt

The hydrous, possibly carbonated, partial melt channel could have originated from decompression melting during mantle upwelling at the mid-ocean ridge. Estimates of melt volatile concentration from mid-ocean ridge basalt (MORB) glasses¹⁹ are in the range 0.10–0.65 wt% melt H₂O, indicating that only the highest MPT that we considered (1,500 °C) is consistent with a ridge origin. Yet this also requires large amounts of melt in the present channel (8.9 ± 1.2 vol%, green distributions in Fig. 3). It is unclear how this much melt remains trapped

at the LAB for an extended period of time. Even if the rigid, overlying lithosphere acts as a permeability barrier, such a large concentration of melt should nucleate dykes and rapidly erode the lithosphere¹⁸.

Alternatively, the elevated water content in the melt predicted by the Monte Carlo process could result from the progressive freezing and thinning of the LAB melt channel as the plate ages²⁰. As melt freezes, the water and carbon dioxide preferentially partition into the remaining melt, enriching it further. The increased melt volatile concentration in turn lowers the freezing point, acting as a negative feedback to further stabilize the remaining melt. Thus a relatively low-fraction melt enriched in volatiles (for example, yellow distributions in Fig. 3) could have arisen from a mantle whose bulk hydration was more in line with estimates from MORB glasses²⁵ (50–250 ppm H₂O). Modelling of mid-ocean-ridge melting²⁶, however, suggests that only at the warmest temperatures we evaluated (1,500 °C MPT) is the amount of water extracted from the solid mantle and incorporated into the partial melt under the ridge sufficient to explain such high water contents in the present melt channel (Extended Data Fig. 5). Yet a 1,500 °C MPT in the present channel does not require elevated water contents but, instead, very high melt fractions, which we already ruled out, given the mechanical instability¹⁸. Similarly, melts generated by small-scale convection²⁷ are derived from the same MORB mantle source as the ridge and would also lack sufficient water and/or carbon dioxide. As a result, a further source is needed to account for the excess volatiles inferred by the MT data at more moderate MPTs. Limiting the temperature range to 1,283–1,351 °C (yellow and purple distributions in Fig. 3 and Extended Data Fig. 4), the 90% credible interval spans 1.4–4.6 vol% melt with 1.5–3.4 wt% H₂O for fixed bulk CO₂ content or 1.0–4.2 vol% melt with 5.9–11.8 wt% CO₂ for fixed bulk H₂O content. Such highly enriched mantle melts are only expected in the transition zone, in mantle plumes and in the mantle wedge above subducting slabs²⁸.

We emphasize that these conclusions depend, in part, on the accuracy of the parameterization used to estimate the solidus depression effect of volatiles (see Methods). For example, if the influence of volatiles on the solidus is larger than estimated, then higher melt fractions with lower melt volatile concentrations would be required to explain the resistivity; a reduction in the influence of volatiles on the solidus would have the opposite effect, and lower melt fractions with higher melt volatile concentrations would be required. As such, improvements in the accuracy of these models will enable better constraints on stable melt fraction and melt volatile content inferred from MT data.

Owing to the increase in melt conductivity with alkali content²⁹, it may be possible to explain the LAB channel with very-low-degree alkali-rich basaltic melts derived from more fertile or metasomatized mantle, without having to invoke a volatile-enriched mantle. Such melts are the source of petit-spots, enigmatic intraplate volcanoes first documented on 130–140-Myr-old Pacific Plate seafloor³⁰. However, we can rule out this possibility for three reasons. First, generating alkaline melts requires a colder geotherm (for example, >40-Myr-old plate) and/or a colder MPT³¹, neither of which are the case here. Second, the melt conductivity must exceed 60 S m⁻¹ to explain the LAB channel with 0.3 vol% melt, yet only nephelinitic melts³² (>20 wt% alkali content) and carbonated melts⁸ (>20 wt% CO₂) have been shown to be this conductive, whereas alkaline basaltic melts should be relatively less conductive, owing to their lower alkali content. Third, several Cocos Plate seamounts adjacent to our survey profile were formed intraplate and have tholeiitic compositions and geochemical signatures that point to partial melting of a mixture of MORB source mantle and Galápagos plume material³³. The origin of these seamounts suggests that plume material may have been entrained beneath the lithosphere when the Cocos Plate was more proximal to the plume head, which—for our survey area—occurred around 15 Myr ago³³.

The resistivity of the LAB channel is anomalously conductive compared with other existing MT observations of oceanic plates (Extended Data Fig. 6). The notable exception is a thin, highly conductive LAB channel observed in the equatorial Atlantic Ocean beneath the 0–30-Myr-old oceanic lithosphere of the African Plate¹⁴. That channel has low resistivity comparable with the Cocos Plate channel, which implies a similar need for volatile-enriched melt, given the young plate age. The nearest known plumes to the Cocos Plate and African Plate LAB channels (the Galápagos and Ascension hotspots, respectively) are on the order of 1,000 km away. If these plumes are responsible for the conductive channels, then the far-field influence of plumes extends well beyond the border of their morphological expression on the seafloor. Alternatively, the conductive channels may mark regions where previously undetected plumes deliver enriched melt to the LAB.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-022-04483-w>.

- Karato, S.-i. On the origin of the asthenosphere. *Earth Planet. Sci. Lett.* **321–322**, 95–103 (2012).
- Kawakatsu, H. & Utada, H. Seismic and electrical signatures of the lithosphere–asthenosphere system of the normal oceanic mantle. *Annu. Rev. Earth Planet. Sci.* **45**, 139–167 (2017).

- Stixrude, L. & Lithgow-Bertelloni, C. Mineralogy and elasticity of the oceanic upper mantle: origin of the low-velocity zone. *J. Geophys. Res.* **110**, B03204 (2005).
- Hansen, L. N., Qi, C. & Warren, J. M. Olivine anisotropy suggests Gutenberg discontinuity is not the base of the lithosphere. *Proc. Natl Acad. Sci. USA* **113**, 10503–10506 (2016).
- Hirth, G. & Kohlstedt, D. Water in the oceanic upper mantle: implications for rheology, melt extraction and the evolution of the lithosphere. *Earth Planet. Sci. Lett.* **144**, 93–108 (1996).
- Holtzman, B. K. Questions on the existence, persistence, and mechanical effects of a very small melt fraction in the asthenosphere. *Geochem. Geophys. Geosyst.* **17**, 470–484 (2016).
- Yoshino, T. & Katsura, T. Electrical conductivity of mantle minerals: role of water in conductivity anomalies. *Annu. Rev. Earth Planet. Sci.* **41**, 605–628 (2013).
- Sifré, D. et al. Electrical conductivity during incipient melting in the oceanic low-velocity zone. *Nature* **508**, 81–85 (2014).
- Naif, S., Key, K., Constable, S. & Evans, R. L. Melt-rich channel observed at the lithosphere–asthenosphere boundary. *Nature* **495**, 356–359 (2013).
- Utada, H. & Baba, K. Estimating the electrical conductivity of the melt phase of a partially molten asthenosphere from seafloor magnetotelluric sounding data. *Phys. Earth Planet. Inter.* **227**, 41–47 (2014).
- Sarafian, E. et al. The electrical structure of the central Pacific upper mantle constrained by the NoMelt experiment. *Geochem. Geophys. Geosyst.* **16**, 1115–1132 (2015).
- Katsura, T., Baba, K., Yoshino, T. & Kogiso, T. Electrical conductivity of the oceanic asthenosphere and its interpretation based on laboratory measurements. *Tectonophysics* **717**, 162–181 (2017).
- Selway, K., O'Donnell, J. P. & Özyaydin, S. Upper mantle melt distribution from petrologically constrained magnetotellurics. *Geochem. Geophys. Geosyst.* **20**, 3328–3346 (2019).
- Wang, S., Constable, S., Rychert, C. A. & Harmon, N. A lithosphere–asthenosphere boundary and partial melt estimated using marine magnetotelluric data at the central Middle Atlantic Ridge. *Geochem. Geophys. Geosyst.* **21**, e2020GC009177 (2020).
- Hirschmann, M. M., Tenner, T., Aubaud, C. & Withers, A. C. Dehydration melting of nominally anhydrous mantle: the primacy of partitioning. *Phys. Earth Planet. Inter.* **176**, 54–68 (2009).
- Hirschmann, M. M. Partial melt in the oceanic low velocity zone. *Phys. Earth Planet. Inter.* **179**, 60–71 (2010).
- Blatter, D., Key, K., Ray, A. & Gustafson, C. Bayesian joint inversion of controlled source electromagnetic and magnetotelluric data to image freshwater aquifer offshore New Jersey. *Geophys. J. Int.* **218**, 1822–1837 (2019).
- Havlin, C., Parmentier, E. M. & Hirth, G. Dike propagation driven by melt accumulation at the lithosphere–asthenosphere boundary. *Earth Planet. Sci. Lett.* **376**, 20–28 (2013).
- Le Voyer, M. et al. Carbon fluxes and primary magma CO₂ contents along the global mid-ocean ridge system. *Geochem. Geophys. Geosyst.* **20**, 1387–1424 (2019).
- Mehouachi, F. & Singh, S. C. Water-rich sublithospheric melt channel in the equatorial Atlantic Ocean. *Nat. Geoscience* **11**, 65–69 (2018).
- Whittington, A., Richet, P. & Holtz, F. Water and the viscosity of depolymerized aluminosilicate melts. *Geochim. Cosmochim. Acta* **64**, 3725–3736 (2000).
- Stein, C. A. & Stein, S. A model for the global variation in oceanic depth and heat flow with lithospheric age. *Nature* **359**, 123–129 (1992).
- Hasterok, D. A heat flow based cooling model for tectonic plates. *Earth Planet. Sci. Lett.* **361**, 34–43 (2013).
- Crosby, A. G. & McKenzie, D. An analysis of young ocean depth, gravity and global residual topography. *Geophys. J. Int.* **178**, 1198–1219 (2009).
- Saal, A. E., Hauri, E., Langmuir, C. H. & Perfit, M. R. Vapour undersaturation in primitive mid-ocean-ridge basalt and the volatile content of Earth's upper mantle. *Nature* **419**, 451–455 (2002).
- Dasgupta, R. et al. Carbon-dioxide-rich silicate melt in the Earth's upper mantle. *Nature* **493**, 211–215 (2013).
- Ballmer, M. D., van Hunen, J., Ito, G., Tackley, P. J. & Bianco, T. A. Non-hotspot volcano chains originating from small-scale sublithospheric convection. *Geophys. Res. Lett.* **34**, L23310 (2007).
- Dixon, J. E., Dixon, T. H., Bell, D. R. & Malservici, R. Lateral variation in upper mantle viscosity: role of water. *Earth Planet. Sci. Lett.* **222**, 451–467 (2004).
- Gaillard, F. Laboratory measurements of electrical conductivity of hydrous and dry silicic melts under pressure. *Earth Planet. Sci. Lett.* **218**, 215–228 (2004).
- Hirano, N. et al. Volcanism in response to plate flexure. *Science* **313**, 1426–1428 (2006).
- Pilet, S. In *The Interdisciplinary Earth: A Volume in Honor of Don L. Anderson* (eds Foulger, G. R., Lustrino, M. & King, S. D.) 281–304 (Geological Society of America, 2015).
- Guo, X. et al. Electrical conductivity of CO₂ and H₂O-bearing nephelinitic melt. *J. Geophys. Res. B Solid Earth* **126**, e2020JB019569 (2021).
- Herbrich, A. et al. Cocos Plate Seamounts offshore NW Costa Rica and SW Nicaragua: implications for large-scale distribution of Galápagos plume material in the upper mantle. *Lithos* **212–215**, 214–230 (2015).

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Methods

Obtaining probability distributions for LAB bulk resistivity and temperature

The Cocos Plate MT data⁹ were inverted using a transdimensional Markov chain Monte Carlo algorithm¹⁷ for 1D resistivity models. Here we modify the algorithm slightly to include a depth-dependent prior distribution such that the upper bound on mantle resistivity is the resistivity of dry olivine at 1,300 °C MPT, while the lower bound is held constant at 0.1 ohm-m. Because our tools rely on 1D models, we carefully selected a subset of data (stations s03, s04, s06, s08, s09, s10 and s11, indicated by red triangles in Extended Data Fig. 1) that is compatible with 1D modelling and least affected by multidimensional conductivity structure (Extended Data Fig. 7). In particular, we chose MT stations that had both low splits between the off-diagonal impedance components and where the model responses of 1D ρ_z profiles extracted from the 2D inversion result of ref. ⁹ beneath each station fit the observed transverse-magnetic (TM) mode data well for all but the shortest periods (corresponding to near-surface heterogeneity). This latter test demonstrates the lack of any notable lateral 2D coast effect distortions in the data considered here. For the mode split calculations, we use the relative difference between the complex Z_{TE} and Z_{TM} impedance components:

$$dZ = \frac{|Z_{TE} - Z_{TM}|}{|Z_{mean}|}, \quad (1)$$

in which

$$Z_{mean} = \frac{Z_{TE} + Z_{TM}}{2}, \quad (2)$$

and the TM phase has been wrapped into the same quadrant as the transverse-electric (TE) mode. Stations with $\text{mean}(dZ) < 0.25$ were assumed to have data suitable for 1D modelling. Further, we only consider data in the TM polarization, as it is less sensitive to off-profile structure than the TE mode. The previous 2D analysis used both TM and TE data to detect a subtle anisotropy in the channel, which was inferred to reflect alignment of the melt into a tube-like geometry parallel to plate motion⁹. Here we focus on the TM mode data, as they are sensitive to the high conductivity direction and thus provide an upper limit on the inferred melt and volatile contents. The full Bayesian posteriors for resistivity as a function of depth (on the basis of more than 2 million samples) for two representative MT sites are shown in Extended Data Fig. 2. The resistive lithosphere is evident down to depths of 40–50 km, below which the mantle becomes abruptly conductive.

This low-resistivity zone is unusually conductive compared with other seafloor MT observations, including the Mantle Electromagnetic and Tomography (MELT) experiment^{34,35}, which also imaged a conductive LAB channel, as well as other studies^{2,11,36} that observed a more resistive asthenosphere without a conductive channel. Only the PI-LAB¹⁴ MT experiment imaged a comparably conductive LAB channel. For a summary of asthenosphere resistivity as a function of plate age inferred from these seafloor MT experiments, see Extended Data Fig. 6.

Estimating a distribution for resistivity from the Bayesian posterior requires determining the depth and thickness of the LAB. Although the depth to the top of conductors is well resolved by MT data, the conductor thickness is not^{37,38}. MT data are, however, highly sensitive to the integrated conductivity (conductance) over a depth interval. We therefore select a depth range, delineated by the white dashed lines in Extended Data Fig. 2a, b, consistent with the conductive anomaly beneath the lithosphere shown by the Bayesian inversion, and compute the conductance over this interval for each model in the combined posterior ensemble for the 1D-compatible MT sites. We then calculate the equivalent conductivity of a single layer spanning the entire depth range by

dividing the conductance of each model by the thickness of the interval. The combined distribution for equivalent resistivity (reciprocal of equivalent conductivity) over the depth range 40–75 km is better constrained than the distributions obtained by simply selecting the resistivity of each model at the midpoint of this depth range. We used these distributions for equivalent resistivity in the Monte Carlo process described below.

The probability distribution for mantle temperature was assumed to be uniform, spanning the endmember thermal models at 57.5 km, the midpoint of the depth interval indicated by the white dashed lines in Extended Data Fig. 2a, b. This yields a temperature uncertainty of 202 °C, spanning the temperature range 1,216–1,418 °C.

Details of the Monte Carlo process

The iterative Monte Carlo process varies slightly, depending on whether bulk hydration or bulk carbon is allowed to vary while the other is held constant. For fixed bulk CO₂, at the temperature drawn from the temperature distribution, mantle hydration is increased (starting from 0 ppm) until the bulk resistivity of the mantle matches the resistivity drawn from its distribution. If this can be achieved without triggering melting, then the melt fraction for this temperature/resistivity combination is recorded as zero. Otherwise, the required melt fraction and its corresponding melt volatile content are recorded. If more than one combination of bulk volatile inventory/melt fraction/melt volatile content is compatible with a given resistivity (for example, small amounts of carbonatite melt versus larger amounts of silicate melt), all possible solutions are recorded. Once this process has been repeated a sufficient number of times, a distribution for these variables can be estimated by computing a normalized histogram.

The process is the same for fixed mantle hydration, except that bulk CO₂ concentration does not affect solid mantle resistivity in the absence of melt. In addition, the effect of carbon on the mantle solidus is constant and does not depend on CO₂ content. This means that, if the sampled temperature is colder than the volatile-influenced solidus (T_{volatile}) and the sampled resistivity is too low to be explained by hydrous mantle, then this combination of resistivity and temperature cannot be explained by variable bulk CO₂ at the chosen fixed mantle hydration. This is noted and new resistivity and temperature samples are drawn.

When allowing bulk mantle H₂O to vary, we fixed bulk mantle CO₂ at 100 ppm. Similarly, when allowing bulk mantle CO₂ to vary, we fixed bulk mantle H₂O at 240 ppm. We chose these values on the basis of regional²⁵ and global¹⁹ estimates of melt CO₂ and melt H₂O concentration in MORB glasses.

Modelling ridge melting

To quantify the amount of water available for a sub-lithospheric melt channel fed from a mid-ocean-ridge source, we modelled instantaneous melting under a mid-ocean ridge from 50–150 km depth using the petrological relationships described below. We assumed that melting above 50 km forms the crust, whereas the melt generated from 50–150 km could be available for a sub-lithospheric melt channel. For temperature, we assumed MPTs of 1,340, 1,400, 1,445 and 1,500 °C (to match the orange, yellow, purple and green distributions in Fig. 3) at a depth of 6 km (base of the crust) and an adiabatic gradient of 0.3 °C km⁻¹ beneath that. We assumed 100 ppm bulk CO₂ and 240 ppm bulk H₂O, as elsewhere in this study. We then summed the total thickness of melt and melt H₂O produced over the depth range 50–150 km (see Extended Data Fig. 5). Because we do not model melt migration or mantle flow, nor do we model batch/fractional melting, these quantities are first-order estimates only, yet sufficient to conclude that at all but the warmest temperatures, the amount of water inferred in the hydrous melt channel exceeds the available water through mid-ocean-ridge melting.

Physical models

Temperature. Temperature is a critical variable affecting melt stability, yet there is notable uncertainty in upper-mantle temperature.

Article

To ensure that we sample an adequate temperature range, we compute temperature as a function of depth using plate-cooling models^{22,23}. Owing to a recent study showing that the solidus of peridotite is probably 60 °C warmer than previously thought³⁹, we explore MPT in the range 1,340–1,500 °C (roughly equivalent to 1,280–1,440 °C without the 60 °C shift). We use the model in ref.²³ with a 1,340 °C MPT and a plate age of 22 Myr as the cold endmember, and the model in ref.²² with a 1,500 °C MPT and plate age of 19 Myr as the warm endmember. Throughout this study, we consider the temperature range 1,216–1,418 °C, which spans these two endmember models at 57.5 km depth, the midpoint of the depth range 40–75 km over which the sub-lithosphere conductive anomaly is best resolved by the Bayesian inversion of the MT data.

Volatiles, the peridotite solidus and partial melting. Partial melting in the upper mantle occurs when the mantle temperature exceeds the peridotite solidus. The amount of partial melt produced is assumed to be a linear function of ΔT (refs.^{16,40}),

$$\Delta T = T - T_s = F \frac{dT}{dF} \quad (3)$$

in which T is the mantle temperature, T_s the volatile-free (dry) peridotite solidus, ΔT is the temperature difference between the mantle temperature and the peridotite solidus, and F is the melt fraction (by weight). $\frac{dT}{dF}$ determines the linear relationship between ΔT and the melt fraction, and is a function of pressure¹⁶,

$$\frac{dT}{dF} = -40P + 450 \quad (4)$$

in which P is pressure (GPa). The dry solidus, T_s , is also modelled to be a function of pressure

$$T_s = aP^2 + bP + c + \Delta \bar{T} \quad (5)$$

in which P is pressure (GPa), a , b and c are regression coefficients⁴¹ and $\Delta \bar{T} = 60^\circ\text{C}$ is the upward shift in the dry peridotite solidus³⁹. We use the following relationship⁴² between depth (km) and pressure (GPa)

$$P = \frac{z - 5}{30.2} \quad (6)$$

In this study, we consider the effect of the addition of the volatiles water and carbon dioxide. These act to depress the peridotite solidus, enabling melt generation at lower mantle temperatures. We calculate the wet peridotite solidus, T_{wet} , using¹⁵

$$T_{\text{wet}} = \frac{T_s}{1 - \frac{R}{M\Delta S} \ln(1 - X_{\text{OH}}^{\text{melt}})} \quad (7)$$

in which $X_{\text{OH}}^{\text{melt}}$ is the mole fraction of dissociated hydroxyl ions (OH^-) in the melt, R is the gas constant, M is the number of grams in one mole of silicate and ΔS is the entropy of fusion per unit mass of peridotite. Here we follow ref.¹⁵ and take ΔS to be $0.4 \text{ J K}^{-1} \text{ g}^{-1}$ and M to be 59 g mol^{-1} . Higher M values can be adopted using other silicate molecular units (for example, units with 4 or 8 oxygens), but 59 g mol^{-1} seems to fit the experimental data best for melt H_2O contents less than 6 wt% (ref.¹⁵). The hydroxyl ion mole fraction is given by (D. Hasterok, personal communication)

$$X_{\text{OH}}^{\text{melt}} = \frac{2M \frac{C_{\text{H}_2\text{O}}^{\text{melt}}}{100}}{18.02 + \frac{C_{\text{H}_2\text{O}}^{\text{melt}}}{100} (2M - 18.02)} \quad (8)$$

in which $C_{\text{H}_2\text{O}}^{\text{melt}}$ is the water concentration in the melt (measured in wt%).

We calculate the peridotite solidus depression owing to carbon dioxide using²⁶

$$\Delta T_s^{\text{CO}_2} = a \times C_{\text{CO}_2}^{\text{melt}} + b \times \ln \left(\frac{100 - c \times C_{\text{CO}_2}^{\text{melt}}}{100} \right) \quad (9)$$

in which $C_{\text{CO}_2}^{\text{melt}}$ is the carbon dioxide concentration in the melt (wt%) and a , b and c are pressure-dependent constants²⁶ (here we use the values for 2 GPa). Combining equations (7) and (9) yields the volatile-influenced solidus, T_{volatile}

$$T_{\text{volatile}} = T_{\text{wet}} - \Delta T_s^{\text{CO}_2} \quad (10)$$

By replacing the dry solidus, T_s , with the volatile-depressed solidus, T_{volatile} , in equation (3), we obtain the following relationship for melt fraction as a function of temperature and melt volatile content

$$F = \frac{T - T_{\text{volatile}}}{dT/dF} \quad (11)$$

The melt volatile concentrations, $C_{\text{H}_2\text{O}}^{\text{melt}}$ and $C_{\text{CO}_2}^{\text{melt}}$, that determine T_{volatile} are functions of their respective bulk volatile inventories, melt fraction and their respective partitioning between the solid and melt phases

$$C_{\text{H}_2\text{O}}^{\text{melt}} = \frac{C_{\text{H}_2\text{O}}^{\text{bulk}}}{D + F(1 - D)} \quad (12)$$

$$C_{\text{CO}_2}^{\text{melt}} = \frac{C_{\text{CO}_2}^{\text{bulk}}}{F} \quad (13)$$

in which $C_{\text{H}_2\text{O}}^{\text{bulk}}$ and $C_{\text{CO}_2}^{\text{bulk}}$ are the bulk water and carbon dioxide concentrations (in wt%), respectively, and D is the water partition coefficient between the solid (peridotite) and melt phases^{15,42}. In the presence of melt, water strongly partitions into the melt phase, whereas carbon dioxide is considered entirely incompatible with upper-mantle minerals and completely partitions into the melt up to a saturation point, which we take to be 38 wt% (ref.²⁶).

There is no closed-form expression for F given its interdependence on $C_{\text{H}_2\text{O}}^{\text{melt}}$ and $C_{\text{CO}_2}^{\text{melt}}$. Instead, F can be solved for numerically by defining a function

$$h(F) = F - \frac{T - T_{\text{volatile}}}{dT/dF} \quad (14)$$

and finding its roots on the interval $[0, 1]$. In the event that no melt is predicted, $C_{\text{CO}_2}^{\text{melt}}$ is assumed to reach saturation (38 wt%).

Calculating electrical resistivity from mantle properties. Mantle electrical resistivity is a complex function of composition, temperature, bulk mantle hydration, melt fraction and melt volatile content⁴³. In this study, we follow ref.⁴⁴ for calculating the sub-solidus mantle resistivity under hydrous conditions. Although independent laboratory studies agree that hydration reduces mantle resistivity, there is notable disagreement on the magnitude of this effect. We considered both the resistive endmember formulation of the Yoshino group^{45,46} and the conductive endmember of the Karato group^{47,48} for olivine and pyroxene resistivity and found no effect on our results: melt is required for the conductive channel in either case. This is because the observed resistivity in the LAB channel is sufficiently low as to require hydration levels that exceed the solidus and therefore require melt.

Basaltic melts are several orders of magnitude less resistive than their source rock^{49,50}. The addition of volatiles (CO₂ and H₂O) further decreases the melt resistivity by around an order of magnitude. We adopt a semi-empirical Arrhenius relationship for the resistivity of basaltic melts with dissolved volatiles from ref.⁸

$$\sigma_{\text{melt}} = \sigma_{\text{H}_2\text{O}} \exp\left(-\frac{\Delta E_{\text{H}_2\text{O}}}{RT}\right) + \sigma_{\text{CO}_2} \exp\left(-\frac{\Delta E_{\text{CO}_2}}{RT}\right) \quad (15)$$

in which R is the gas constant, $\sigma_{\text{H}_2\text{O}}$ and σ_{CO_2} are pre-exponential factors and $\Delta E_{\text{H}_2\text{O}}$ and ΔE_{CO_2} are activation energies. The latter are empirical functions of the water and carbon dioxide concentrations

$$\Delta E_{\text{volatile}} = a \exp(-b C_{\text{volatile}}) + c \quad (16)$$

$$\sigma_{\text{volatile}} = \exp(d \Delta E_{\text{volatile}} + e) \quad (17)$$

In the above, a , b , c , d and e are constants and C_{volatile} is the melt H₂O or CO₂ content in wt%. Carbon-rich melts have an extremely low resistivity and can be notably less resistive than even water-rich melts. For the effect on resistivity to be pronounced, however, the carbon concentration in the melt must be elevated⁸. An alternative empirical formulation that considers only the effect of water on basaltic melt resistivity is⁵¹

$$\log(\sigma_{\text{melt}}) = 2.172 - \frac{860.82 - 204.46 \sqrt{C_{\text{H}_2\text{O}}}}{T - 1,146.8} \quad (18)$$

in which $C_{\text{H}_2\text{O}}$ is the concentration of water in the melt in wt% and T is the temperature in kelvin. This relationship was derived using data from laboratory experiments with up to 6.3 wt% H₂O. We compared equation (15) with equation (18) for a range of water and carbon concentrations and found that carbon has little effect on the resistivity of the melt below about 6 wt% CO₂ (Extended Data Fig. 8). The two formulations diverge beyond about 6 wt% H₂O. This is not of serious concern here, as our analysis fortuitously does not require melt H₂O concentrations higher than this. Because we consider the combined effects of both water and carbon dioxide, we use the formulation in equation (15) throughout this analysis.

Basaltic melt is so much less resistive than the host rock that the bulk electrical resistivity of a two-phase mantle system (solid plus melt) is strongly dependent on the interconnectedness of the melt phase. Because partial melt can be fully interconnected even at very small vol% (ref.⁵²), we use the well-known Hashin–Shtrikman model upper bound (HS+) to calculate the bulk resistivity of two-phase mantle

$$\sigma_{\text{bulk}}^{\text{HS+}} = \sigma_{\text{melt}} \frac{1}{1 - \frac{3\phi_{\text{solid}}(\sigma_{\text{melt}} - \sigma_{\text{solid}})}{3\sigma_{\text{melt}} - \phi_{\text{melt}}(\sigma_{\text{melt}} - \sigma_{\text{solid}})}} \quad (19)$$

in which ϕ_{melt} and ϕ_{solid} are the volume fractions and σ_{melt} and σ_{solid} the conductivities of the melt and solid phases, respectively. The conversion between mass fraction and volume fraction is given by⁸

$$\phi_{\text{melt}} = \left(1 + \left(\frac{1}{X_{\text{m}}^{\text{melt}}} - 1\right) \frac{d_{\text{melt}}}{d_{\text{peridotite}}}\right)^{-1} \quad (20)$$

in which $X_{\text{m}}^{\text{melt}}$ is the mass fraction of melt and d_{melt} and $d_{\text{peridotite}}$ are the densities of silicate melt and peridotite, respectively.

The effects of temperature, bulk mantle hydration, melt fraction and melt H₂O content on bulk mantle resistivity are shown in Extended Data Fig. 9. We assume a mantle composition of 60 vol% olivine and 40 vol% pyroxene, which affects the solid mantle resistivity, as well as the partitioning of water between the solid and melt phases. The curves in Extended Data Fig. 9a are truncated where the solidus is reached (for

a given temperature, at which further bulk hydration would trigger melting). The resistivity curves in Extended Data Fig. 9b were calculated assuming a dry solid mantle.

Estimating mantle properties from electrical resistivity and temperature. As shown in Extended Data Fig. 9, a given resistivity can be explained by various combinations of temperature, mineral hydration, partial melt and melt volatile content. Estimating these mantle properties from bulk electrical resistivity alone is therefore a non-unique process. If partial melt is present, it will dominate the bulk resistivity, whereas the influence of temperature and hydration of the solid matrix on resistivity will be secondary. Hence a reduction in bulk resistivity can primarily be explained by either increasing the melt fraction (equation (19)) or reducing the resistivity of the melt phase by means of the addition of volatiles (equation (15)). This trade-off is illustrated by the orange curves in Fig. 1. This ambiguity can be resolved by adding the requirement that the melt phase be in thermodynamic equilibrium with the surrounding solid mantle.

The amount of melt present in a parcel of mantle at thermodynamic equilibrium is determined by the difference between its temperature and the solidus (equation (3)). The solidus, in turn, is depressed by the bulk volatile concentration of the parcel (equation (10)). Thus as the volatile concentration of a parcel of mantle increases at constant temperature, the melt fraction increases. And because volatiles partition preferentially into the melt phase, the melt volatile concentration increases as well. This is illustrated qualitatively by the blue curves in Fig. 1 and shown numerically in Extended Data Fig. 10a for a few discrete temperatures. Extended Data Figure 10 examines this trade-off for water only, with bulk CO₂ concentration set to zero. The melt fraction/melt H₂O content curves begin at the point where the temperature of the parcel exceeds its solidus (that is, when $T > T_{\text{volatile}}$).

Because simultaneously increasing the melt fraction and melt volatile concentration of a parcel of mantle reduces its bulk resistivity, the stable melt fraction and volatile content of a parcel of mantle can be constrained by bulk resistivity measurements. Extended Data Figure 10b shows the bulk resistivity of a parcel of mantle for the stable combinations of melt fraction and H₂O concentration shown in Extended Data Fig. 10a. Combinations of melt fraction and melt H₂O content that produce a constant bulk resistivity (irrespective of petrological stability) are shown in Extended Data Fig. 10c for a few discrete resistivity values (coloured curves). The constant-resistivity curves in Extended Data Fig. 10c were calculated at $T = 1,326^\circ\text{C}$ and assume a dry solid matrix, as the water partitions overwhelmingly into the melt phase. The petrologically stable combinations of melt fraction and melt H₂O concentration from Extended Data Fig. 10a are also plotted (black curves).

Comparing Fig. 1 and Extended Data Fig. 10c, it is clear that, if both resistivity and temperature were precisely known (for example, 3 ohm-m and $1,326^\circ\text{C}$), the stable melt fraction and melt H₂O concentration (and, by implication, total mantle hydration) would be given by the intersection of the relevant constant-resistivity (colour) and constant-temperature (black) curves (for example, the blue dot in Extended Data Fig. 10d). If temperature were specified and quantitative uncertainty bounds on resistivity were known (for example, from Bayesian inversion of electromagnetic data), then the petrologically stable combinations of melt fraction and melt H₂O concentration would be constrained to lie along a portion of the relevant constant-temperature curve (for example, the green curve in Extended Data Fig. 10d). If, instead, resistivity were specified and uncertainty bounds placed on temperature, the petrologically stable combinations of melt fraction and melt H₂O concentration would be constrained to lie along a portion of the relevant constant-resistivity curve (for example, red curve in Extended Data Fig. 10d). If uncertainty were specified in both temperature and resistivity, the petrologically stable combinations would lie within a 2D region (for example, the grey region in Extended Data Fig. 10d).

Data Availability

The magnetotelluric data that were inverted and analysed in this study can be found at: <https://doi.org/10.5281/zenodo.5510673>. Source data are provided with this paper.

Code Availability

The transdimensional Bayesian inversion code is available at <https://bayesian1dem.bitbucket.io/>. The suite of codes used to model the partial melting processes as well as the bulk resistivity of two-phase mantle will be released on request. Source data are provided with this paper.

34. Evans, R. L. et al. Geophysical evidence from the MELT area for compositional controls on oceanic plates. *Nature* **437**, 249–252 (2005).
35. Baba, K., Chave, A. D., Evans, R. L., Hirth, G. & Mackie, R. L. Mantle dynamics beneath the East Pacific Rise at 17°S: insights from the Mantle Electromagnetic and Tomography (MELT) experiment. *J. Geophys. Res.* **111**, B02101 (2006).
36. Baba, K. et al. Electrical conductivity imaging of the Philippine Sea upper mantle using seafloor magnetotelluric data. *Phys. Earth Planet. Inter.* **183**, 44–62 (2010).
37. Parker, R. L. The magnetotelluric inverse problem. *Geophys. Surv.* **6**, 5–25 (1983).
38. Medin, A. E., Parker, R. L. & Constable, S. Making sound inferences from geomagnetic sounding. *Phys. Earth Planet. Inter.* **160**, 51–59 (2007).
39. Sarafian, E., Gaetani, G. A., Hauri, E. H. & Sarafian, A. R. Experimental constraints on the damp peridotite solidus and oceanic mantle potential temperature. *Science* **355**, 942–945 (2017).
40. Katz, R. F., Spiegelman, M. & Langmuir, C. H. A new parameterization of hydrous mantle melting. *Geochem. Geophys. Geosyst.* **4**, 1073 (2003).
41. Hirschmann, M. M. Mantle solidus: experimental constraints and the effects of peridotite composition. *Geochem. Geophys. Geosyst.* **1**, 2000GC000070 (2000).
42. O’Leary, J. A., Gaetani, G. A. & Hauri, E. H. The effect of tetrahedral Al³⁺ on the partitioning of water between clinopyroxene and silicate melt. *Earth Planet. Sci. Lett.* **297**, 111–120 (2010).
43. Naif, S., Selway, K., Murphy, B. S., Egbert, G. & Pommier, A. Electrical conductivity of the lithosphere-asthenosphere system. *Phys. Earth Planet. Inter.* **313**, 106661 (2021).
44. Naif, S. An upper bound on the electrical conductivity of hydrated oceanic mantle at the onset of dehydration melting. *Earth Planet. Sci. Lett.* **482**, 357–366 (2018).
45. Yoshino, T., Matsuzaki, T., Shatskiy, A. & Katsura, T. The effect of water on the electrical conductivity of olivine aggregates and its implications for the electrical structure of the upper mantle. *Earth Planet. Sci. Lett.* **288**, 291–300 (2009).
46. Zhang, B. et al. Electrical conductivity of enstatite as a function of water content: implications for the electrical structure in the upper mantle. *Earth Planet. Sci. Lett.* **357–358**, 11–20 (2012).
47. Wang, D., Mookherjee, M., Xu, Y. & Karato, S.-i. The effect of water on the electrical conductivity of olivine. *Nature* **443**, 977–980 (2006).
48. Dai, L. & Karato, S.-i. Electrical conductivity of orthopyroxene: implications for the water content of the asthenosphere. *Proc. Jpn. Acad. B* **85**, 466–475 (2009).
49. Presnall, D. C., Simmons, C. L. & Porath, H. Changes in electrical conductivity of a synthetic basalt during melting. *J. Geophys. Res. Solid Earth* **77**, 5665–5672 (1972).
50. Tyburczy, J. A. & Waff, H. S. Electrical conductivity of molten basalt and andesite to 25 kilobars pressure: geophysical significance and implications for charge transport and melt structure. *J. Geophys. Res. Solid Earth* **88**, 2413–2430 (1983).
51. Ni, H., Keppler, H. & Behrens, H. Electrical conductivity of hydrous basaltic melts: implications for partial melting in the upper mantle. *Contrib. Mineral. Petrol.* **162**, 637–650 (2011).
52. Gardés, E., Laumonier, M., Massuyeau, M. & Gaillard, F. Unravelling partial melt distribution in the oceanic low velocity zone. *Earth Planet. Sci. Lett.* **540**, 116242 (2020).

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Author contributions S.N. and K.K. collected and processed the magnetotelluric data. D.B. and S.N. wrote the partial melting and bulk resistivity modelling codes. A.R. and D.B. wrote the Bayesian inversion codes. D.B., S.N. and K.K. modelled the magnetotelluric data and produced the figures. All authors contributed to writing the manuscript.

Competing interests The authors declare no competing interests.

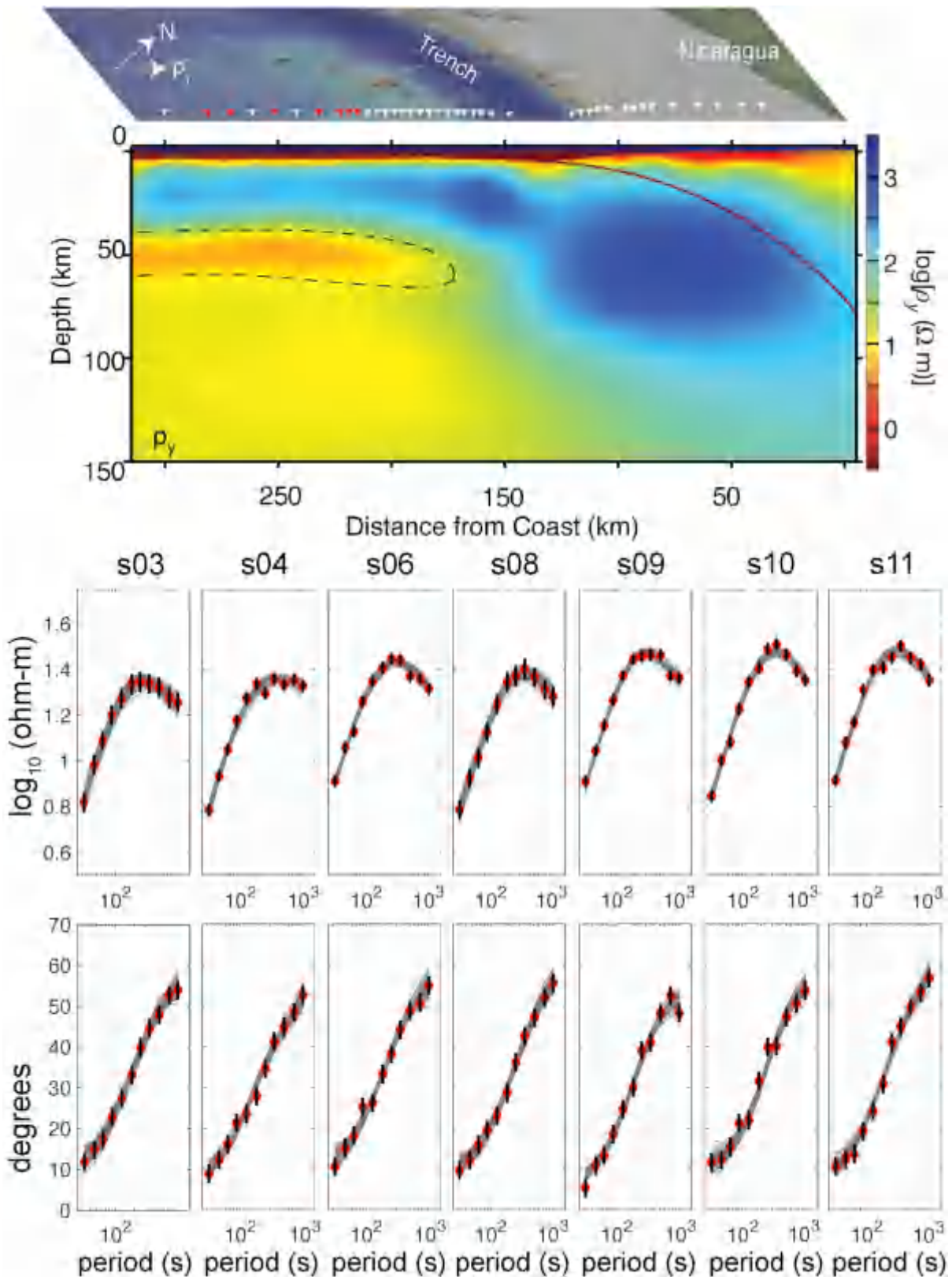
Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41586-022-04483-w>.

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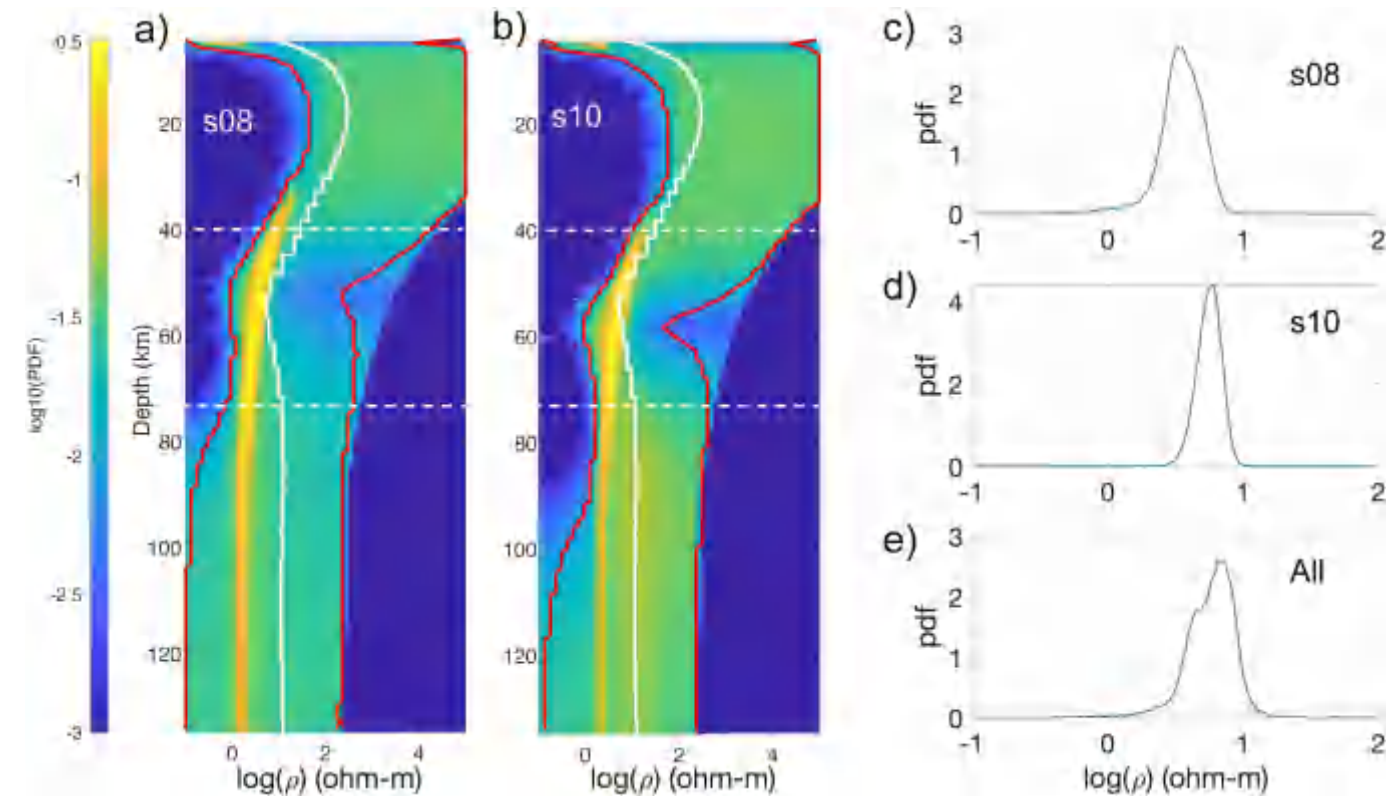
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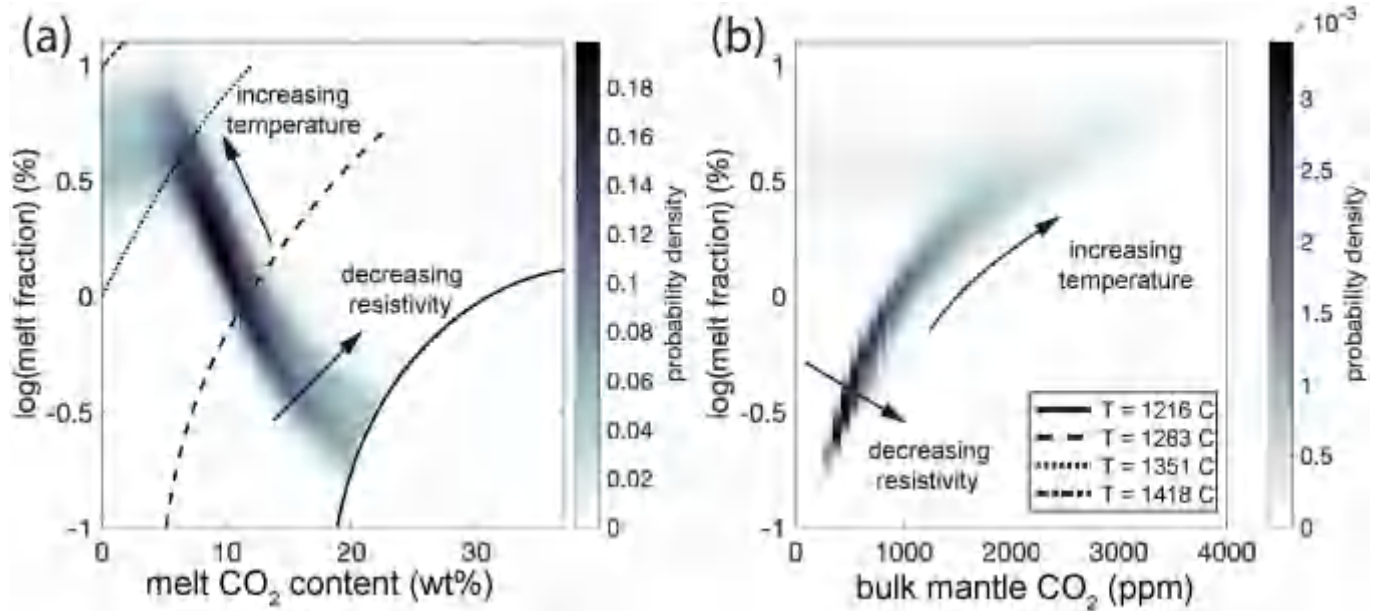
Extended Data Fig. 1 | Survey map, resistivity model estimate and field data from the Cocos Plate MT experiment offshore Nicaragua⁹. Top, survey map and 2D deterministic resistivity model estimate, modified from ref. ⁹. The subducting Cocos Plate boundary is shown in dark red. Warm colours are more conductive, cool colours resistive. The conductive anomaly at the LAB is

outlined by the black dashed line. MT stations are shown as triangles, with those stations included in this study indicated in red. Bottom, field MT data from all MT sites used in this study shown as apparent resistivity (upper row) and phase (lower row). A representative subset of Bayesian inversion model responses are shown in grey lines.



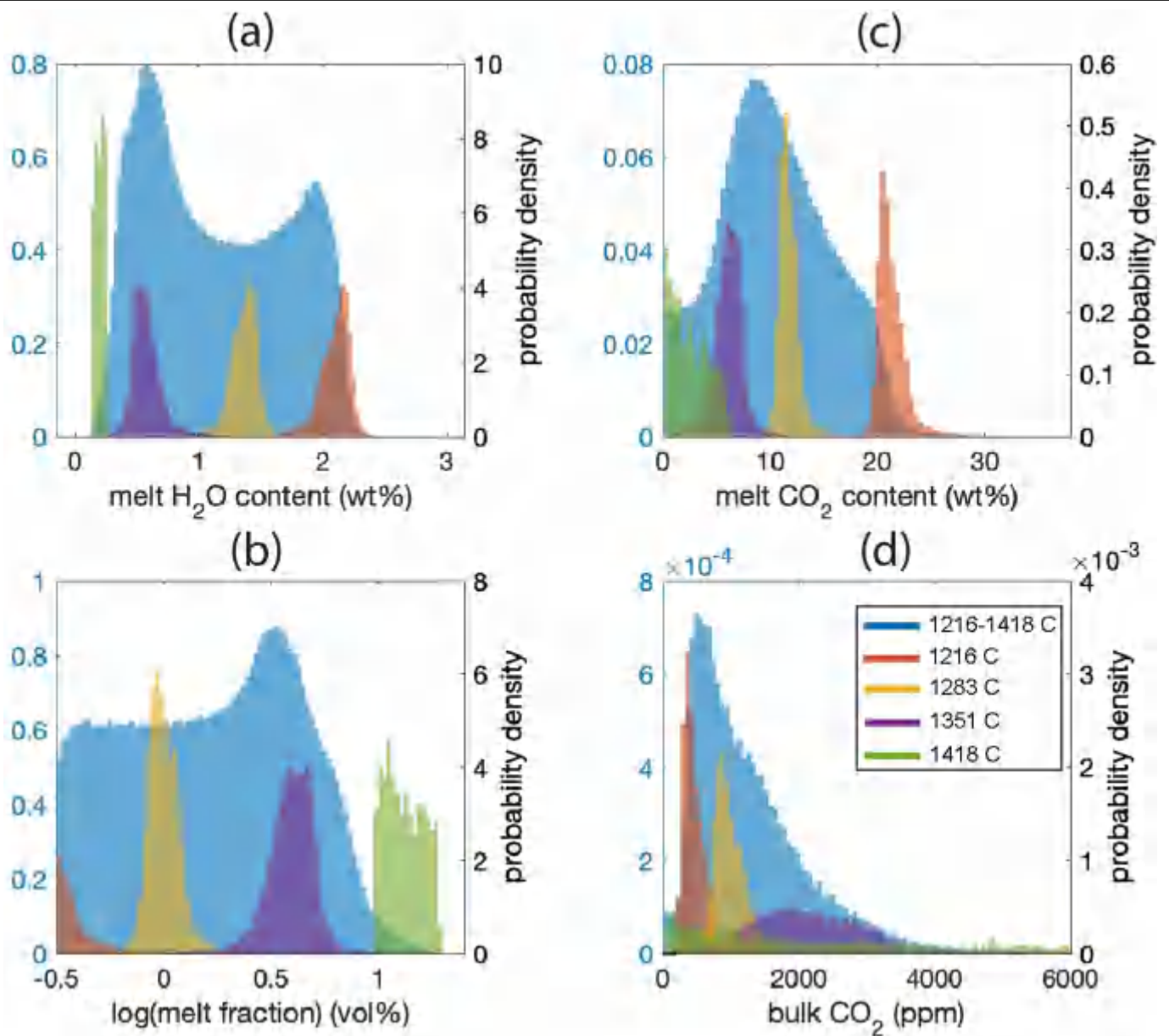
Extended Data Fig. 2 | Posterior probability density distributions for resistivity. a, b, Marginal probability as a function of depth for MT sites s08 and s10, respectively. Warm colours indicate regions of higher probability. Red lines denote the 5th and 95th percentiles of the distribution at each depth, respectively. The solid white line is a vertical profile through the 2D deterministic inversion result (Extended Data Fig. 1) taken at each MT site. One-dimensional marginal distributions for resistivity are obtained for each

MT sounding used in this study by computing the conductance of each model in the posterior ensemble over the interval between 40 km and 75 km (white dashed lines), then normalizing by the thickness of the interval (equivalent resistivity). **c–e,** Equivalent resistivity distributions for MT sites s08, s10 and the combined distribution for all MT sites. The distribution in **e** is used throughout this analysis.



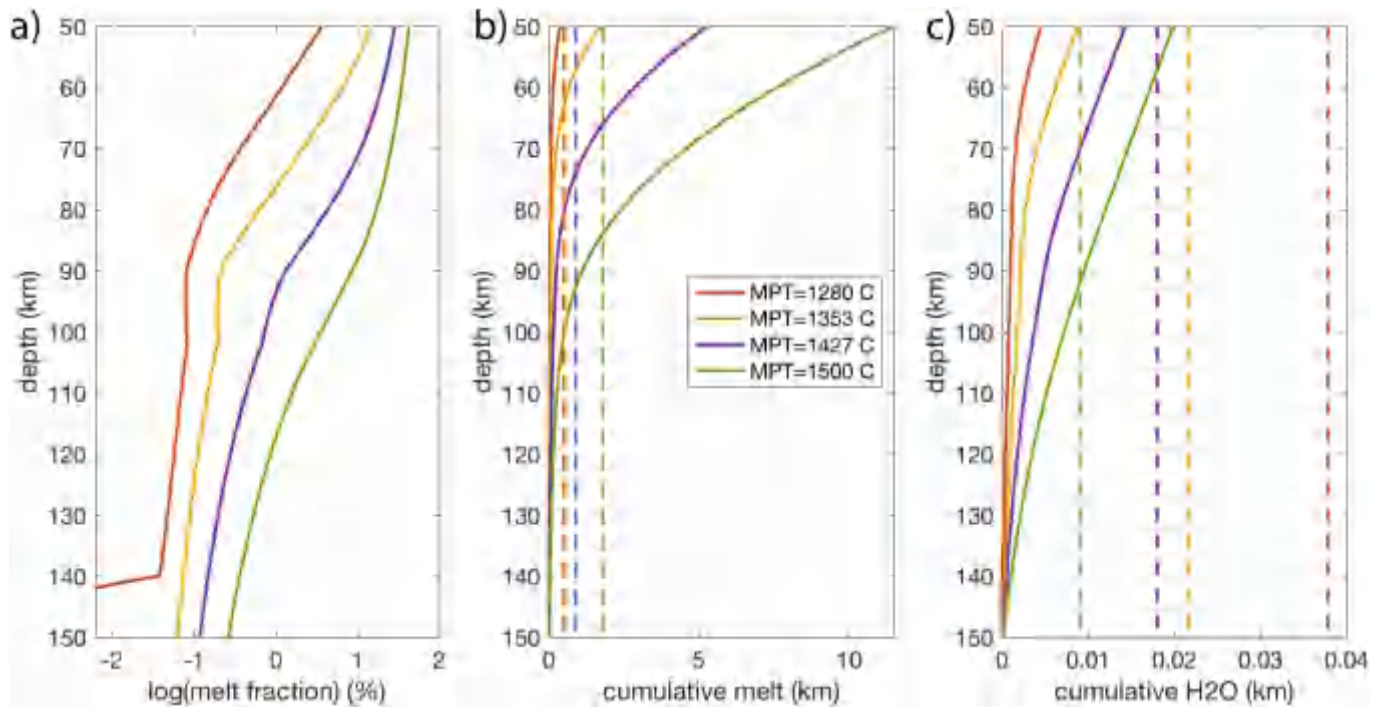
Extended Data Fig. 3 | Probability distributions for melt fraction and melt CO₂ content (a) or bulk CO₂ (b), for the combined MT soundings, estimated from a Monte Carlo method. Dashed lines indicate isotherms. Bulk water was

held constant at 240 ppm. Nearly all the (T, ρ) draws require large melt fraction and/or high bulk CO₂ concentration.



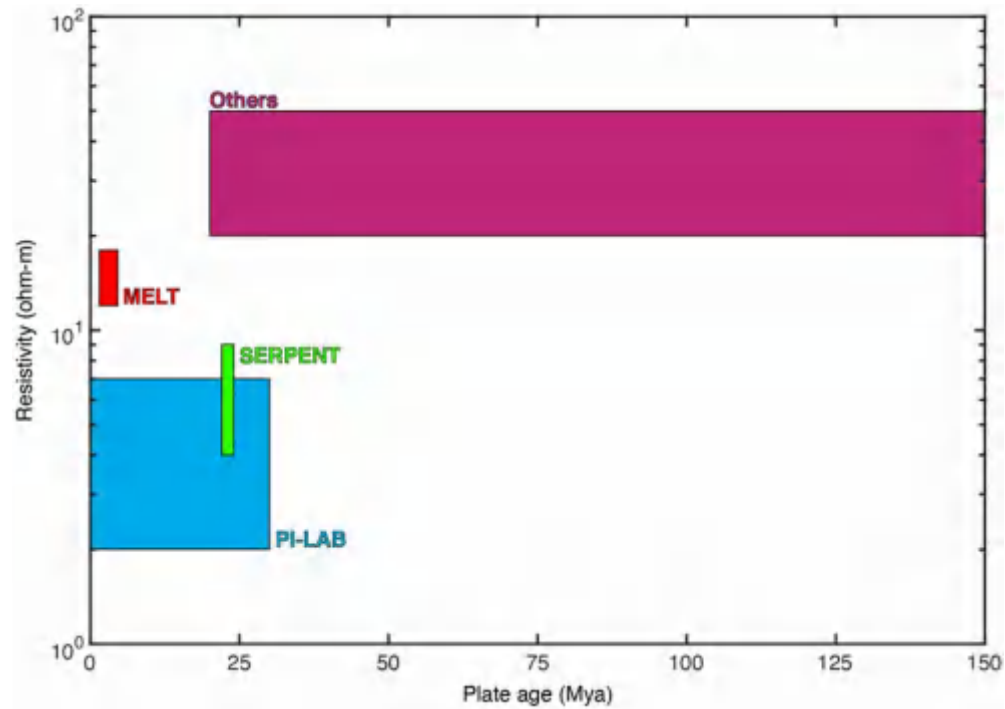
Extended Data Fig. 4 | Marginal distributions for melt H_2O (a), melt fraction (b), melt CO_2 (c) and bulk hydration (d), similar to Fig. 3 but with bulk hydration held constant at 240 ppm. Only at the coldest temperatures are low-degree, high- CO_2 -concentration melts stable. At warmer temperatures,

bulk carbon concentrations must be elevated to match bulk resistivity. At the warmest temperatures, melt fractions are high enough that bulk resistivity is nearly insensitive to bulk carbon concentration.



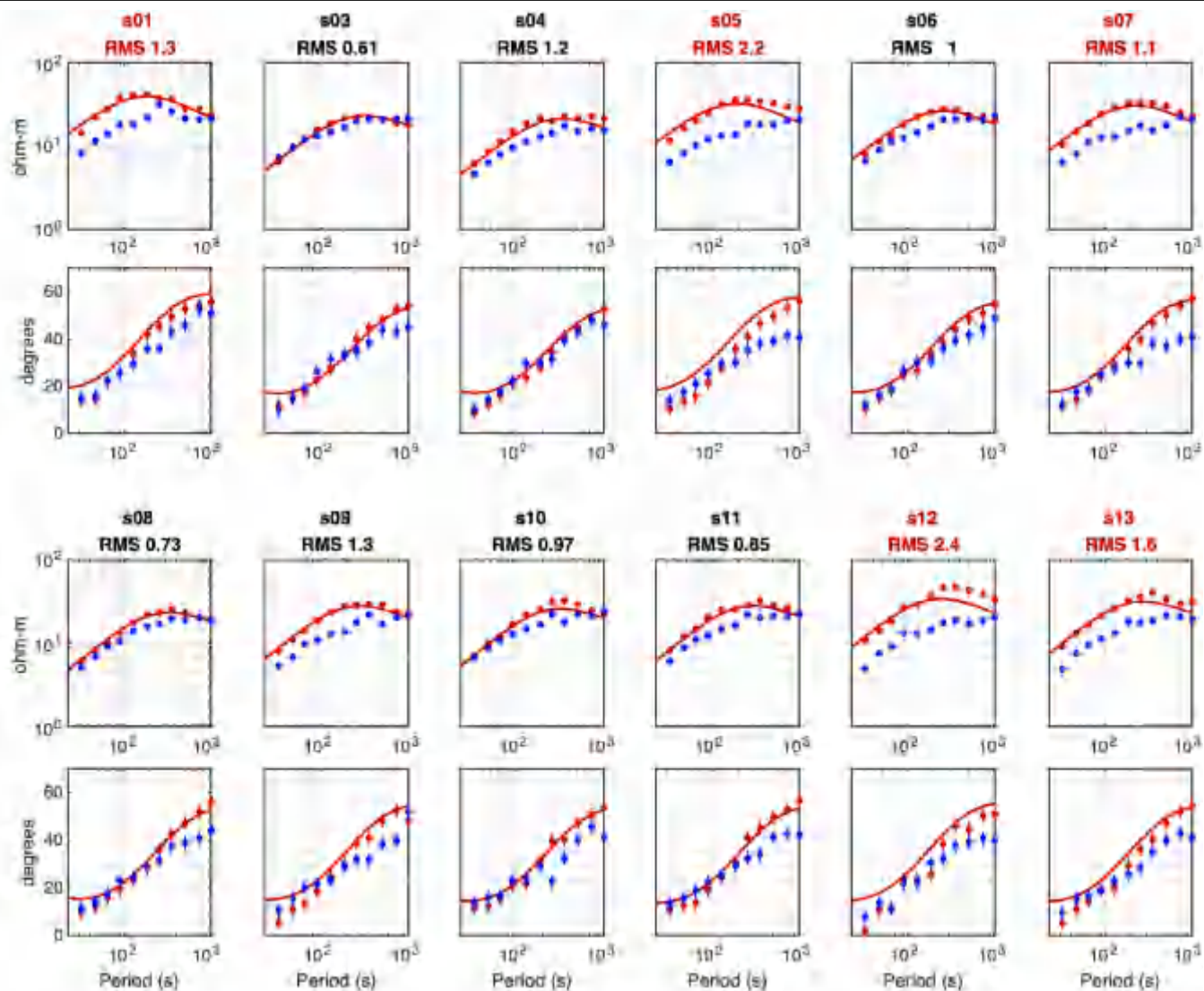
Extended Data Fig. 5 | Melt and dissolved water produced as a function of depth beneath a mid-ocean ridge for various mantle potential temperatures. **a**, Melt fraction as a function of depth. Cumulative kilometres of melt (starting from a depth of 150 km) (**b**) and the corresponding cumulative water in the melt (**c**). Dashed lines are the MT estimated melt and water

contents of the hydrous channel; mantle potential temperatures and colours match Fig. 3. Only at the warmest temperatures is the water extracted from the solid mantle and incorporated into the partial melt under the ridge greater than the amount of water inferred in the hydrous channel.

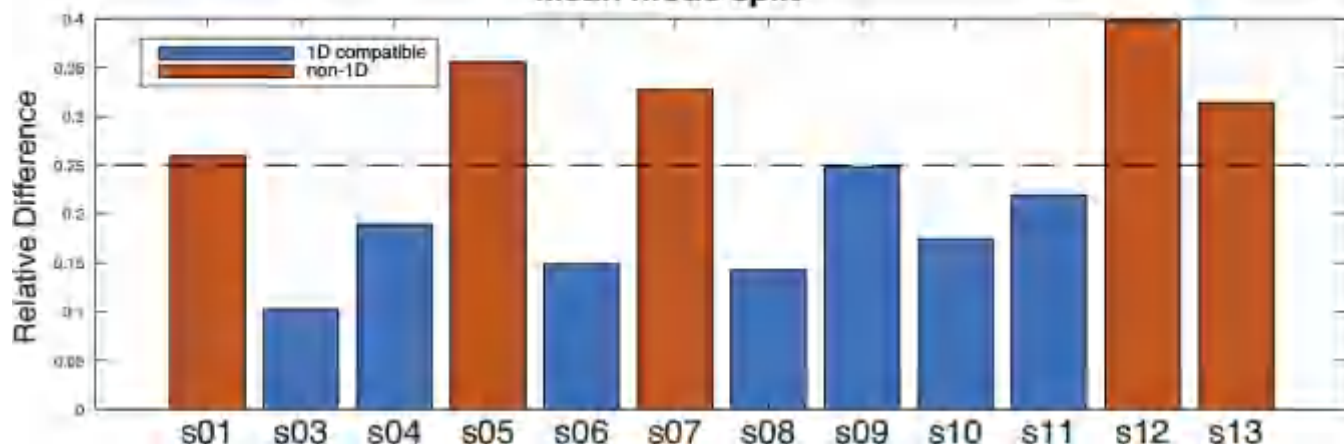


Extended Data Fig. 6 | Comparison of asthenosphere resistivity from marine MT observations of oceanic plates. Conductive LAB channels were observed in the MELT^{34,35}, SERPENT⁹ and PI-LAB¹⁴ experiments. The magenta region indicates other MT studies^{2,11,36} that observed more resistive

asthenosphere with no indications of conductive channels. For anisotropic inversion models^{9,34,35}, only the inline resistivity (TM mode) is shown. Values are approximate.



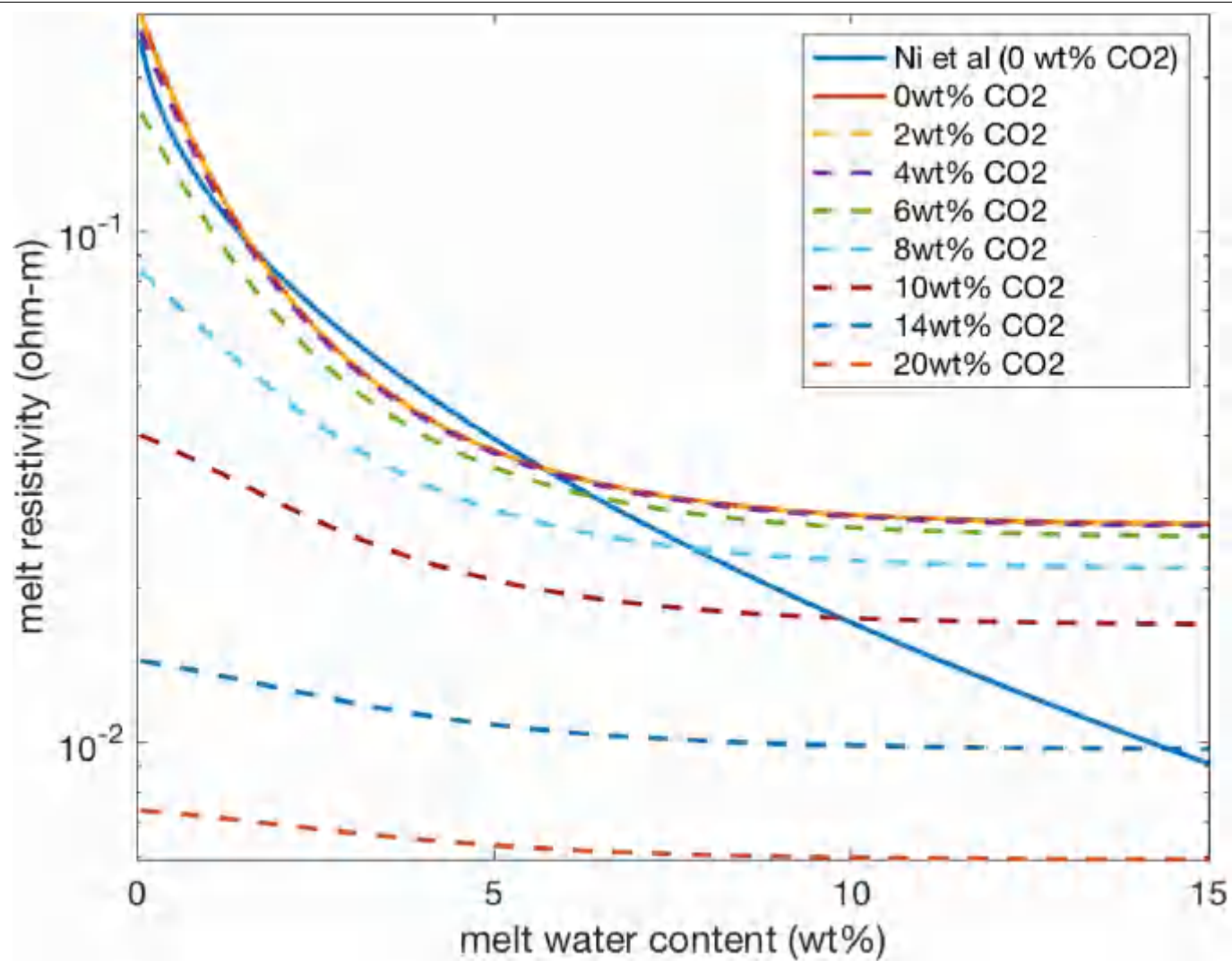
Mean mode split



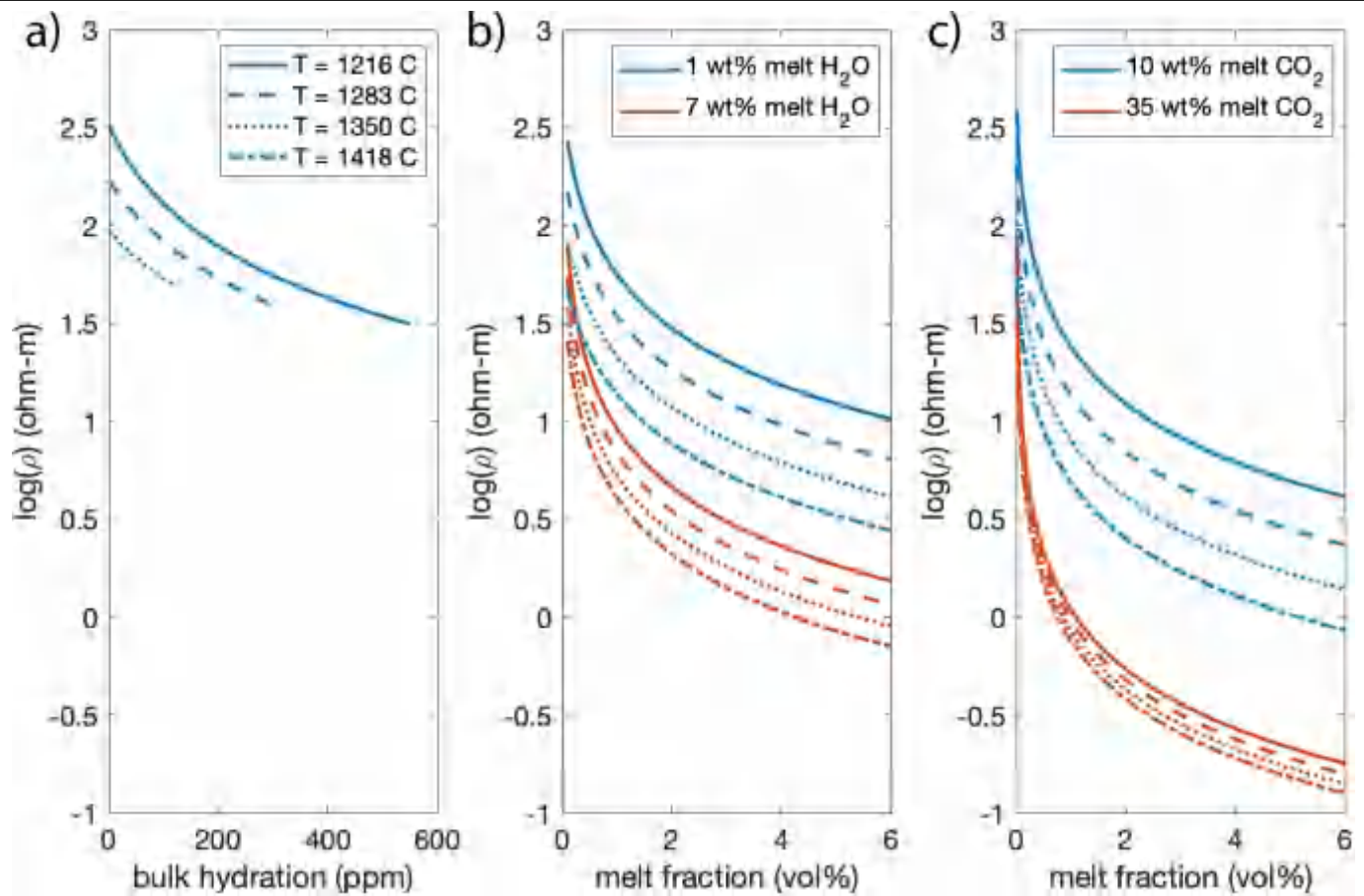
Extended Data Fig. 7 | Suitability of the observed MT data for 1D modelling.

Top, TE mode (blue circles) and TM mode (red circles) data for all stations are shown with model responses (red lines) for 1D ρ_r resistivity profiles extracted from the 2D inversion model of ref. ⁹ beneath each station. Small differences at the shortest periods are owing to lateral variations in sediment thickness and shallow crustal structure. The RMS fit between the TM data and 1D profile responses are given for each station. Low RMS values indicate data that may be

suitable for 1D modelling. RMS values shown in red text indicate stations deemed incompatible with 1D modelling owing to either large RMS values or large mean mode splits. Bottom, selection of 1D compatible data is further refined by examining the mean mode split between the TE and the TM complex impedances at each station. Mean mode splits with a relative difference below 0.25 indicate stations with data compatible with 1D interpretation.

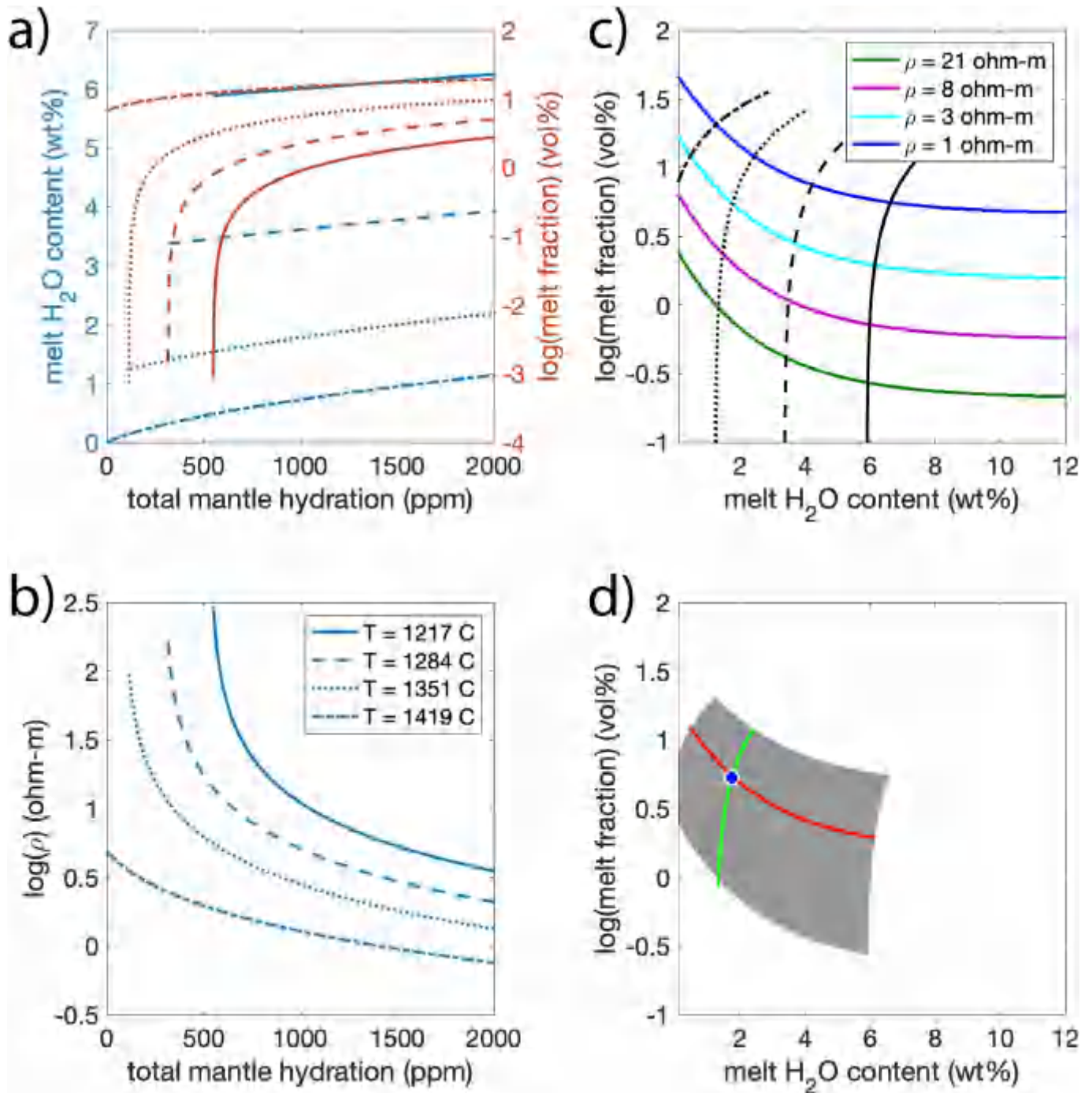


Extended Data Fig. 8 | Effect of CO₂ on resistivity of hydrous melt at $T = 1,400$ °C (ref. ⁵¹). Shown for comparison is the relationship for purely hydrous melts⁵¹ (solid blue line). The effect of carbon on melt resistivity is not pronounced until the carbon concentration in the melt exceeds about 6 wt%.



Extended Data Fig. 9 | Effect of temperature, bulk hydration, melt fraction and melt volatile content on mantle bulk electrical resistivity. a, For a damp mantle with no melt, both temperature and bulk hydration reduce bulk resistivity. The line styles in all three plots follow the temperatures in **a**. The curves in **a** are truncated where the solidus is reached and melt would be

produced. **b,** A dry mantle with hydrous melt is considerably less resistive, even for small melt water concentrations. **c,** Carbon dioxide has a strong influence on melt resistivity, but only at high melt CO_2 concentrations. Bulk resistivity was computed without carbon dioxide in **b** and without water in **c**. Mantle composition was assumed to be 60% olivine, 40% pyroxene.



Extended Data Fig. 10 | Using bulk resistivity and temperature to constrain petrologically stable melt fraction and melt H₂O concentration. For stable melts, melt fraction and melt water concentration both increase as a function of total mantle hydration (a), exerting the dominant control on bulk resistivity (b). Constant-resistivity combinations of melt fraction and melt water content (c, coloured curves) are plotted alongside the stable, constant-temperature combinations from a (c, black curves). For known T and ρ , the stable melt

fraction and melt water content are known precisely (d, blue dot). If uncertainty in T or ρ is included, the stable combinations of melt fraction and melt water content plot along a line (d, red or green lines, respectively). If both T and ρ are uncertain, the petrologically stable combinations lie in a 2D region (d, grey region). Dashed lines (whether blue, orange or black) indicate curves at constant temperature throughout the figure.